

A Hybrid Regularization Framework for Correlated Feature Selection in High-Dimensional Data

Scaled Group Lasso (SGLASSO)

Bahadır Yüzbaşı, PhD

Joint work with Jiguo Cao

Department of Econometrics

İnönü University, Türkiye

HDDA-XV, Istanbul Medipol University

12 August 2026

Problem and method

1. Why grouped high-dimensional predictors are difficult under collinearity.
2. How SGLASSO modifies the usual group-penalty shrinkage target.
3. How the block-coordinate solver fits the path efficiently.

Evidence and interpretation

1. Simulation evidence on prediction, recovery, selection, and runtime.
2. GenAtHum gene-expression application with tissue/cell-type groups.
3. Practical take-home messages for grouped data analysis.

1. Motivation

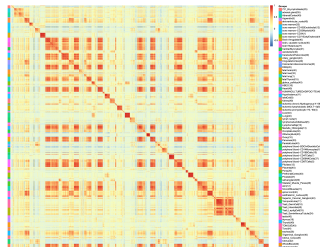
Known groups, correlated predictors, and a prediction–sparsity trade-off

Why correlated groups make selection difficult

High-dimensional predictors often arrive with a known group structure: pathways, basis expansions, tissues, or related sensors.

- **Featurewise selection** can fragment a meaningful group.
- **Groupwise selection** respects structure, but correlation makes overlapping groups hard to separate.
- **Prediction tuning** can retain related groups when they improve out-of-sample error.

Question: can group sparsity and stable estimation coexist under collinearity?



What changes in grouped high-dimensional data?

Structure

Predictors are not isolated: genes, tissues, pathways, basis functions, and repeated measurements naturally form groups.

Collinearity

Groups can carry similar signal. Removing one correlated group may improve sparsity but hurt prediction or coefficient recovery.

Tuning

Validation-based tuning often favors prediction, so the selected model may be broader than the sparsest scientifically plausible model.

DESIGN GOAL

SGLASSO is designed as a flexible regularization point between pure group sparsity and shrinkage toward a data-informed target.

2. Scaled Group Lasso

A groupwise penalty with a data-informed shrinkage target

A grouped linear model

$$\mathbf{y} = \beta_0 \mathbf{1} + \sum_{j=1}^J \mathbf{X}_j \boldsymbol{\beta}_j + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}).$$

What is selected?

Each $\boldsymbol{\beta}_j \in \mathbb{R}^{p_j}$ is a block of coefficients. A groupwise method either keeps or removes the block as a unit.

Why is it difficult?

When $\mathbf{X}_j$ and $\mathbf{X}_\ell$ are correlated, several groups can carry overlapping signal. Aggressive selection may discard useful information.

Group Lasso is the starting point

$$\hat{\beta}^{\text{GL}} = \arg \min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \sum_{j=1}^J w_j \|\beta_j\|_2 \right\}, \quad w_j = \sqrt{M_j}.$$

Key idea

Select or remove a whole coefficient block.

Why it matters

Scientific grouping is used directly rather than ignored.

Open issue

All selected groups are shrunk toward zero, even when groups are correlated.

SGLASSO retains Group Lasso's selection rule and changes the shrinkage target for correlated groups [Yuan and Lin, 2006].

The modeling move: change the shrinkage target

Method	Group sparsity	Quadratic shrinkage target
Group Lasso	Yes	None
Group elastic net	Yes	0
SGLASSO	Yes	$d\hat{\beta}_j^{\text{LSE}}$

Interpretation

The target preserves group selection while modifying the shrinkage destination. The parameter d controls its influence [Zou and Hastie, 2005, Helwig, 2025, James et al., 1961, Arashi et al., 2021].

Why a scaled target can help

Zero-centered shrinkage

Group Lasso and group elastic net pull selected coefficients toward zero. This is effective for sparsity, but can be too aggressive when correlated groups share signal.

SGLASSO shrinkage

The quadratic part pulls group j toward $d\hat{\beta}_j^{\text{LSE}}$. This keeps the groupwise selection mechanism but allows a nonzero shrinkage target.

ROLE OF d

The scale parameter does not create a new grouping structure; it changes how strongly the selected group is pulled toward the least-squares-based direction.

SGLASSO combines group sparsity and scaled shrinkage

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \alpha \sum_{j=1}^J w_j \|\boldsymbol{\beta}_j\|_2 + \frac{\lambda(1-\alpha)}{2} \sum_{j=1}^J w_j \|\boldsymbol{\beta}_j - d \hat{\boldsymbol{\beta}}_j^{\text{LSE}}\|_2^2 \right\}, \quad w_j = \sqrt{M_j},$$

λ

Overall regularization level.

α

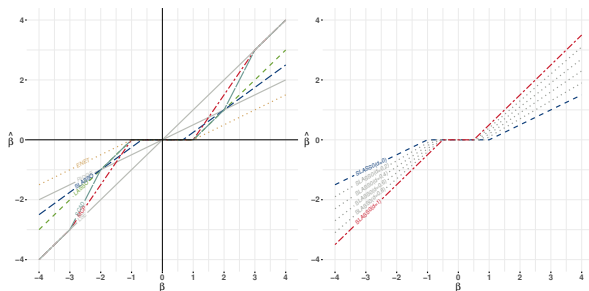
Balances group sparsity and quadratic shrinkage.

d

Sets the scaled least-squares target.

Here M_j is the number of predictors in group j . When $\alpha = 1$, the quadratic term vanishes and SGLASSO reduces to group lasso.

The scale parameter changes the shrinkage target



Interpretation

Quadratic shrinkage targets $d\hat{\beta}_j^{\text{LSE}}$, not only zero.

$d = 0$: zero-centered quadratic

shrinkage. $0 < d < 1$:

an intermediate, scaled target. $d = 1$:

the full group-LSE target.

From the objective to one block subproblem

After standardization and group orthonormalization, hold all blocks except j fixed. Let

$$z_j = \frac{1}{n} \mathbf{X}_j^\top \mathbf{r} + \boldsymbol{\beta}_j, \quad A_j = z_j + \lambda_2 w_j d \widehat{\boldsymbol{\beta}}_j^{\text{LSE}}, \quad \lambda_1 = \lambda \alpha, \quad \lambda_2 = \lambda(1 - \alpha).$$

The block update minimizes, up to a constant,

$$\frac{1 + \lambda_2 w_j}{2} \left\| \boldsymbol{\beta}_j - \frac{A_j}{1 + \lambda_2 w_j} \right\|_2^2 + \lambda_1 w_j \|\boldsymbol{\beta}_j\|_2.$$

What enters A_j ?

The partial residual score, the current block coefficient, and the scaled least-squares target.

Why this is useful

The multivariate block problem reduces to one proximal group-thresholding operation.

This is the groupwise analogue of a coordinate-descent subproblem [Yuan and Lin, 2006, Breheny and Huang, 2015].

KKT conditions give two groupwise cases

The blockwise KKT condition is

$$\mathbf{0} \in (1 + \lambda_2 w_j) \widehat{\boldsymbol{\beta}}_j - \mathbf{A}_j + \lambda_1 w_j \partial \left\| \widehat{\boldsymbol{\beta}}_j \right\|_2.$$

Case 1: inactive group

At $\widehat{\boldsymbol{\beta}}_j = \mathbf{0}$, the subgradient has norm at most one. Therefore,

$$\left\| \mathbf{A}_j \right\|_2 \leq \lambda_1 w_j \quad \Rightarrow \quad \widehat{\boldsymbol{\beta}}_j = \mathbf{0}.$$

Case 2: active group

When $\left\| \mathbf{A}_j \right\|_2 > \lambda_1 w_j$, the solution is aligned with $\mathbf{A}_j$ and is shrunk as a vector.

Group soft-thresholding is the BCD update

Define the vector soft-thresholding operator

$$\mathcal{S}_\tau(\mathbf{a}) = \left(1 - \frac{\tau}{\|\mathbf{a}\|_2}\right)_+ \mathbf{a}, \quad \mathbf{a} \in \mathbb{R}^{M_j}.$$

Then the exact update is

$$\widehat{\boldsymbol{\beta}}_j = \frac{1}{1 + \lambda_2 w_j} \mathcal{S}_{\lambda_1 w_j}(\mathbf{A}_j).$$

Selection and shrinkage are distinct

The operator selects the whole group; the denominator applies quadratic shrinkage afterward.

Residual update

Use $\mathbf{r} \leftarrow \mathbf{r} - \mathbf{X}_j(\boldsymbol{\beta}_j^{\text{new}} - \boldsymbol{\beta}_j)$ after each block. No full $\mathbf{X}\boldsymbol{\beta}$ recomputation is needed.

The formula remains valid for unequal group sizes through $w_j = \sqrt{M_j}$.

Pathwise SSR: screen first, verify exactly

At the previous path point $\lambda^{(k)}$, SGLASSO uses the score

$$G_j^{(k)} = \left\| \frac{1}{n} \mathbf{X}_j^\top \mathbf{r}(\lambda^{(k)}) + \lambda^{(k)}(1 - \alpha)w_j \left\{ d\widehat{\boldsymbol{\beta}}_j^{\text{LSE}} - \widehat{\boldsymbol{\beta}}_j(\lambda^{(k)}) \right\} \right\|_2.$$

SSR discards a currently inactive group for $\lambda^{(k+1)}$ when

$$G_j^{(k)} < \alpha w_j \left(2\lambda^{(k+1)} - \lambda^{(k)} \right).$$

Fast candidate reduction

Fit from large to small λ , warm-starting from the preceding solution and updating only the retained working set.

KKT verification

SSR is strong, not safe. Check $G_j(\lambda^{(k+1)}) \leq \alpha w_j \lambda^{(k+1)}$; restore and re-fit any violation.

The shifted term is specific to SGLASSO; final KKT checks recover the exact fitted path [Friedman et al., 2007, Tibshirani et al., 2012].

Preparation

1. Standardize predictors.
2. Orthonormalize each group.
3. Build a decreasing λ path.
4. Initialize with the previous path solution.

Path fitting

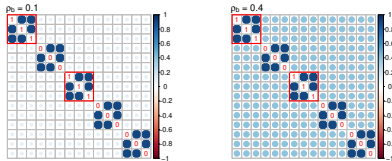
1. Apply SSR to remove unlikely inactive groups.
2. Cycle through active/working groups by BCD.
3. Update residuals incrementally.
4. Check final KKT conditions and refit if needed.

Computational message: reduce repeated full-design calculations while retaining final KKT verification.

3. Simulation evidence

Prediction, recovery, selection, and computational scaling

Simulation design: a controlled correlation experiment



Grouped Gaussian design

HD settings: $n = 100$, $p = 1000$, $J = 200$

Five predictors per group; $s \in \{10, 20, 40\}$ active groups.

Within-group correlation: $\rho_w = 0.9$;
between-group correlation: $\rho_b \in \{0, 0.1, \dots, 0.9\}$.

SNR: 0.05 to 6; tuning is by validation prediction error.

Signal patterns

Homogeneous: common active magnitude.

Weak/strong mixed: two active magnitudes.

Grouped BCD-type estimators

- GLASSO: group sparsity.
- GSCAD and GMCP: nonconvex group penalties.
- GENET: group elastic-net regularization.
- SGLASSO: group sparsity plus scaled shrinkage.

Specialized baseline

ADELIE is included as a modern high-speed grouped-regularization benchmark.

Fair comparison principle

Methods use common training, validation, and test samples.

Read the simulation results as a multi-criterion comparison

Prediction and estimation

Test MSE. Out-of-sample squared prediction error.

Coefficient MSE. Squared error in recovering β .

PVE. Proportion of signal variation explained.

Selection and computation

FDR. False discoveries among selected terms.

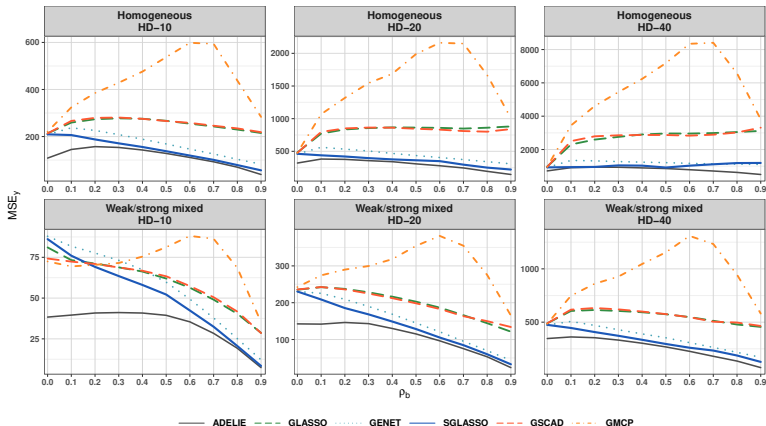
$\hat{s}$. Number of selected groups.

Runtime. Serial fitting time over a regularization path.

INTERPRETATION PRINCIPLE

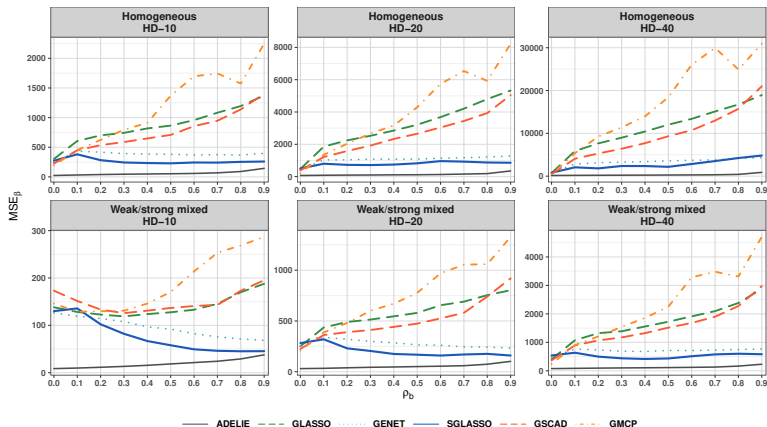
No single criterion fully characterizes a grouped estimator: validation-MSE tuning can favor a broader model if correlated groups improve prediction.

Prediction under correlation



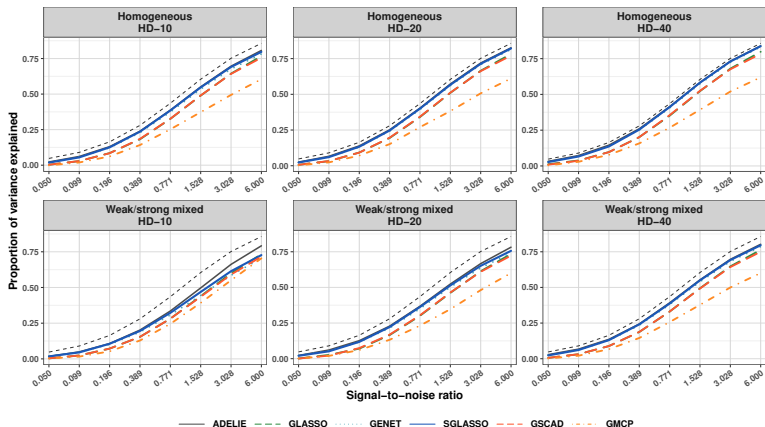
HD: $p = 1000$, $J = 200$, $SNR = 1.528$. SGLASSO remains close to the best predictive methods as between-group correlation increases.

Coefficient recovery under correlation



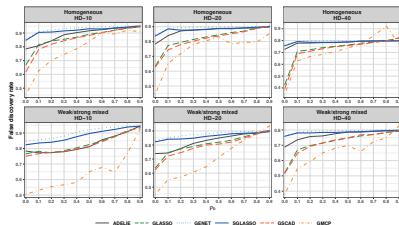
Coefficient MSE asks whether the fitted path recovers the grouped signal, not only whether it predicts well.

PVE improves with signal strength

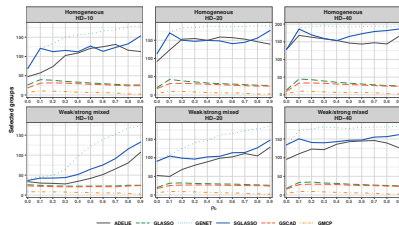


PVE increases with SNR. The dashed curve is $\text{SNR}/(1 + \text{SNR})$; SGLASSO tracks the leading predictive methods.

Selection: prediction versus sparsity



False discovery rate



Average number of selected groups

SGLASSO does not target the smallest possible model. With validation-error tuning, it can select additional correlated groups when doing so improves predictive performance; FDR should therefore be interpreted jointly with test MSE and PVE.

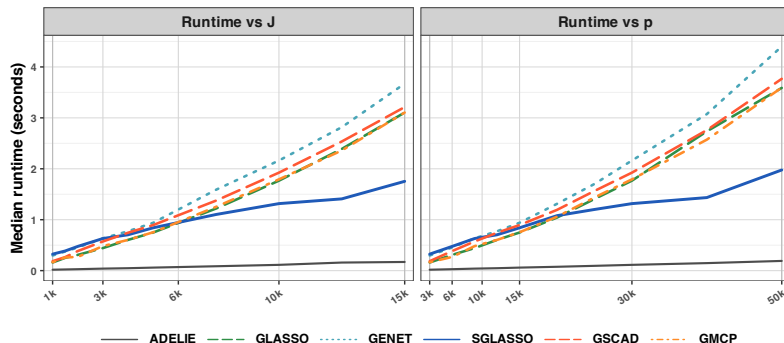
What is stable

- Prediction improves with stronger signal.
- The effect of between-group correlation is visible across all methods.
- SGLASSO is consistently competitive among grouped-penalty estimators.

What is the trade-off

- A smaller selected model is not always the best predictive model.
- Nonconvex group penalties can be sparse, but may lose prediction accuracy.
- SGLASSO often occupies an intermediate prediction-selection operating point.

Scalability among BCD solvers



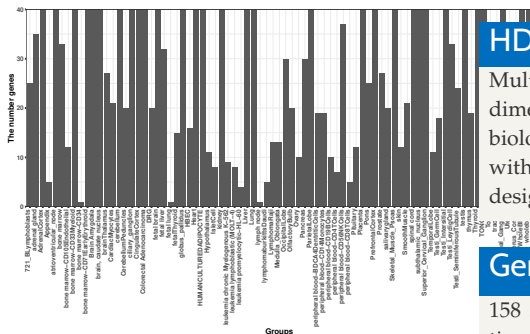
Median serial runtime over 50 replications. ADELIE is the fastest specialized solver; SGLASSO scales favorably among BCD-type grouped-penalty methods.

Helwig [2025], Yang and Hastie [2024].

4. Real-data application

Gene-expression prediction with tissue and cell-type groups

From HDDA to GenAtHum: a grouped omics problem



HDDA/MODAS link

Multi-omics data combine high-dimensional measurements and biological groupings. MODAS within SABITA supports this design-to-analysis workflow.

GenAtHum design

158 samples; 2,045 predictors; 79 tissue/cell-type groups; sizes 1–40. Response: MORF4L1 expression, a gene linked to DNA repair, cell-cycle, and transcriptional regulation.

MODAS | SABITA, Istanbul Medipol University

Why this real-data example is useful

Statistical difficulty

- p is much larger than the number of samples.
- Predictors are grouped by biological tissue or cell type.
- Correlation appears both within and across groups.

Scientific interpretation

- A selected group can correspond to a tissue/cell-type signal.
- Model size matters for interpretability.
- Prediction accuracy matters because the target is gene-expression prediction.

Analysis question: can a grouped penalty improve prediction while preserving interpretable group-level structure?

GenAtHum: prediction and model size

Method	Test MSE (SE)	Relative MSE	$\hat{s}$	$\hat{p}$
ADELIE	0.1302 (0.0068)	0.944	33	436.5
SGLASSO	0.1380 (0.0307)	1.000	25	418.0
GLASSO	0.1644 (0.0321)	1.191	12	89.0
GENET	0.1652 (0.0299)	1.197	13	105.0
GSCAD	0.1732 (0.0211)	1.255	10	84.5
GMCP	0.1891 (0.1043)	1.371	5	28.0

Protocol

Median over 100 common 70/30 train/test splits; each training set uses 5-fold CV.

Reading the table

SGLASSO has the second-smallest test MSE. Its larger selected model is consistent with prediction-focused tuning under correlated groups.

Prediction

ADELIE gives the smallest median test MSE, while SGLASSO gives the second-smallest value and improves over GLASSO, GENET, GSCAD, and GMCP in this application.

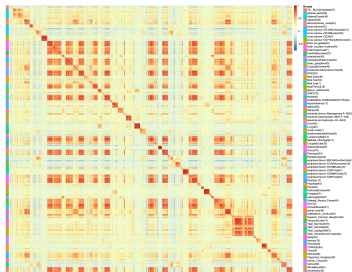
Selection

SGLASSO selects more predictors than GLASSO, GSCAD, and GMCP. This is not unexpected: the tuning criterion is prediction-oriented and the groups are correlated.

PRACTICAL READING

For the GenAtHum data, SGLASSO behaves as a prediction-focused grouped estimator: it is not the sparsest model, but it preserves group structure and gives strong test performance.

GenAtHum: prediction with correlated gene groups



Evaluation protocol

100 repeated 70/30 train/test splits; identical splits and 5-fold CV for every method.

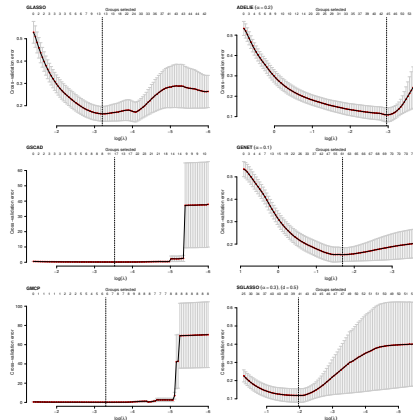
SGLASSO result

Median test MSE: 0.1380. Median model: 25 groups and 418 predictors.

INTERPRETATION

Second-smallest test MSE after ADELIE; lower MSE than GLASSO, GENET, GSCAD, and GMCP.

Cross-validation paths



Full-data 5-fold CV across six methods. SGLASSO can retain a broader correlated model when this improves CV prediction.

5. Take-home message

A flexible groupwise operating point for correlated high-dimensional data

What SGLASSO contributes

1. **A flexible groupwise target:** combines selection with scaled shrinkage toward a least-squares-based target.
2. **A practical solver:** BCD, warm starts, sequential screening, residual updates, and KKT verification.
3. **A complementary operating point:** competitive prediction and estimation under correlated groups, with sparsity controlled by the tuning criterion.

TAKE-HOME MESSAGE

For overlapping or heterogeneous correlated signal, the best predictive model need not be the smallest one. SGLASSO tunes that trade-off.

When would I use SGLASSO?

Use it when

- groups are scientifically meaningful;
- predictors are correlated within or across groups;
- prediction and group-level interpretation both matter;
- zero-centered shrinkage may be too restrictive.

Interpret results as

- a selected set of predictive groups;
- a regularized coefficient path;
- a trade-off among test error, coefficient recovery, and sparsity;
- not merely a minimum-FDR procedure.

Limitations and next steps

Current scope

- Linear Gaussian grouped regression.
- Known, non-overlapping groups.
- Tuning by validation, information criteria, or cross-validation.

Natural extensions

- Generalized linear models.
- Overlapping or hierarchical groups.
- Faster second-order block updates for very large grouped designs.

Closing point: SGLASSO adds a flexible grouped-regularization option when collinearity makes the prediction-selection trade-off visible.

Research setting

- Developed during postdoctoral research at Simon Fraser University, Canada.
- Supported by the TÜBİTAK BİDEB 2219 programme.

Computing and dissemination

- Large simulation experiments were run on the TÜBİTAK ULAKBİM TRUBA infrastructure.
- Published in *The American Statistician* (2026).

RESOURCES

Journal site: *The American Statistician*

GitHub: **sglasso**

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Thank you

Package and reproducible code

<https://github.com/byuzbasi/sglasso>

Contact

Bahadır Yüzbaşı
İnönü University